Introduction to Psychology 1
Cognitive Psychology – lecture 1

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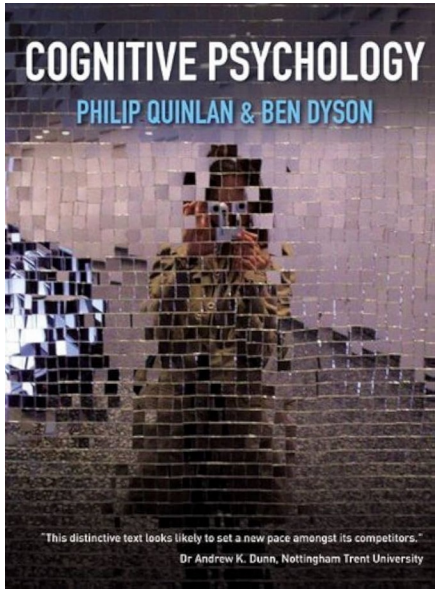
Your learning tools

- **Text book**

  Quinlan & Dyson, Cognitive Psychology

  Eysenk & Keane, Cognitive Psychology a Students Handbook



- **Lecture slides**

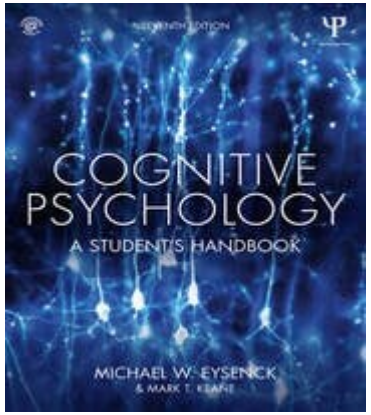
  Blue boxes: **Suggested** further reading. No exam question will be on these readings

- **In-class activities –** Poll Everywhere

- **Test yourself slides**

  Learning outcome slides

  Describe and interpret the evidence slides



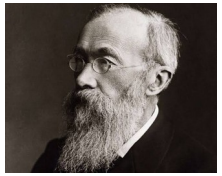
Cognitive Psychology Block – weeks 16 to 20

What's on the menu?

1. What is Cognitive Psychology?
2. Selective Attention
3. Early & Late Selection
4. Resource Theories of Attention
5. Mental Imagery
6. Problem Solving
7. Creativity
8. Decision making
9. Revision lecture for class test
10. Class test

Let's go back in time…

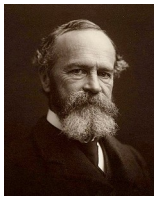
1879



Wilhelm Wundt & Structuralism

The Father of Psychology

1890

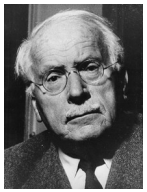


William James & Functionalism

1900's-50's

Watson, Skinner & Behaviourism:

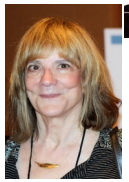
Koffka, Kohler & the Gestalt Approach



Freud, Adler, Jung & the Psychodynamic approach

1950's- 90's

Cognitive Psychology: Information Processing Approach

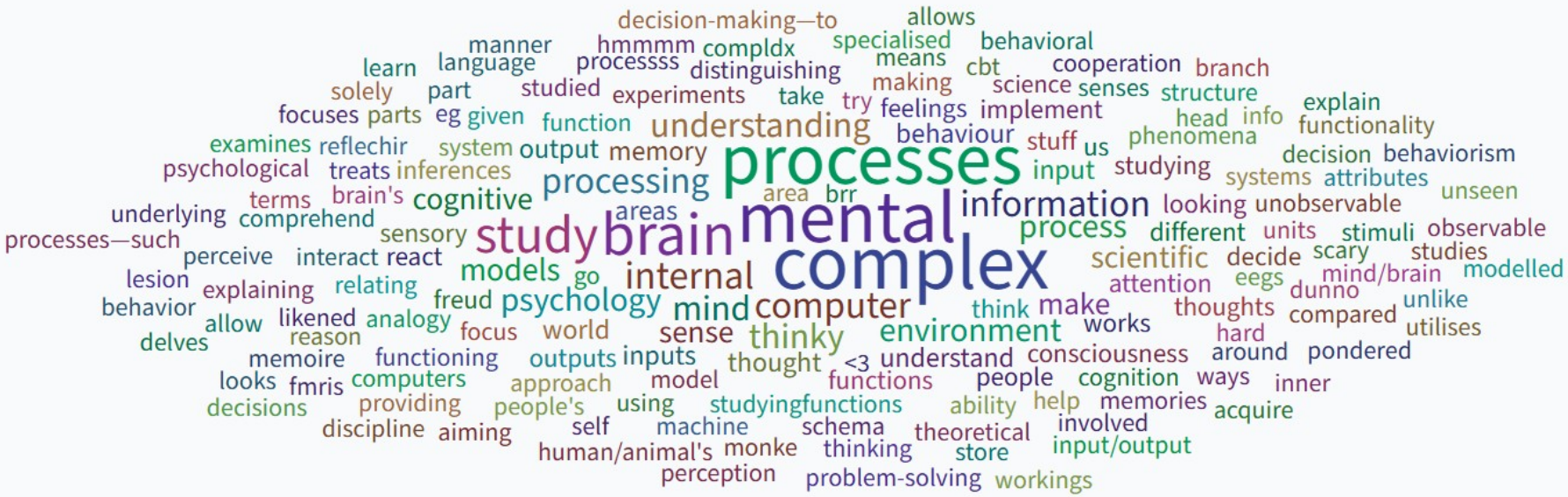


2000….

Cognitive Neuroscience

What is Cognitive Psychology?

What is cognitive psychology?

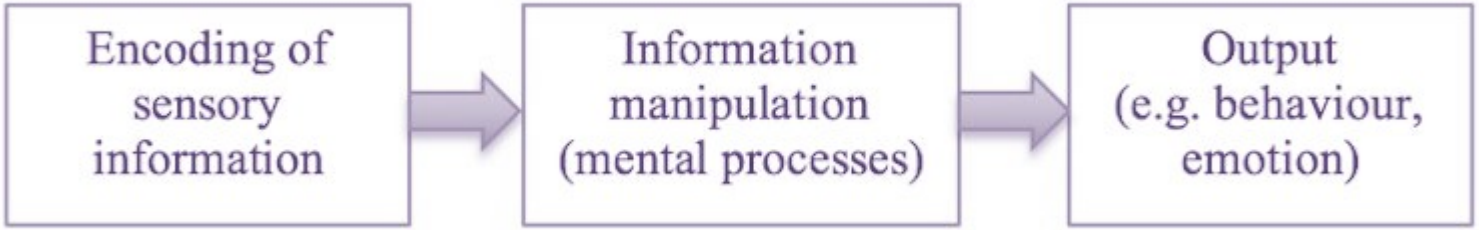


Powered by Poll Everywhere

What is Cognitive Psychology?

- Understand the mental processes that allow us to make sense of our environment, decide how react to the environment and implement those decisions.

- Generate descriptions of how these mental processes function: typically a flow chart



- What about the brain?

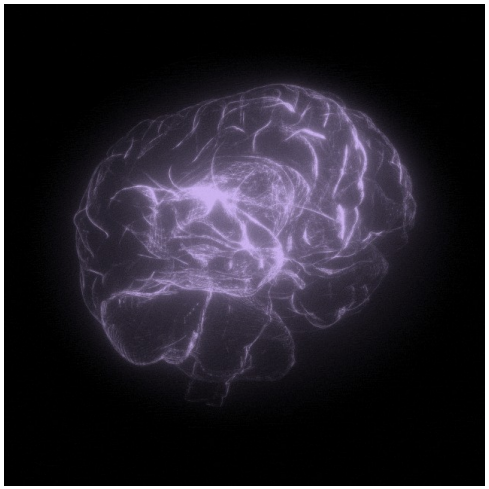
- Useful to understand the brain, but understanding the brain doesn't necessarily produce insights into the mind…

The Mind/Body problem

Type Identity Theory

A mental state is equivalent to a <u>specific</u> pattern of neural events

2 brains problem (Searle 1994)



Functionalism

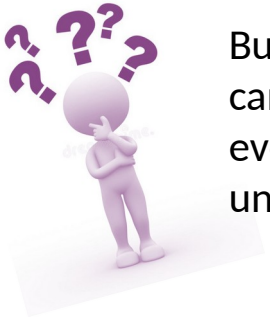
Draws a distinction between

1. Structure of a mental state (the neural activity)
2. Function of a mental state (the consequences of the mental state, e.g. behaviours or new mental states)

Cognitive psychology is about developing a <u>functional</u> explanation of mental processes.

Token Identity Theory

A mental state maps onto variety of different neural events



But if Token theory is true, can knowledge of neural events ever help us understand mental events?

Information Processing Analogy

Human cognition can be conceptualised as a flow of information or ‘signals’

If cognition is the processing of signals, psychologists can measure cognition by measuring signal processing, e.g.

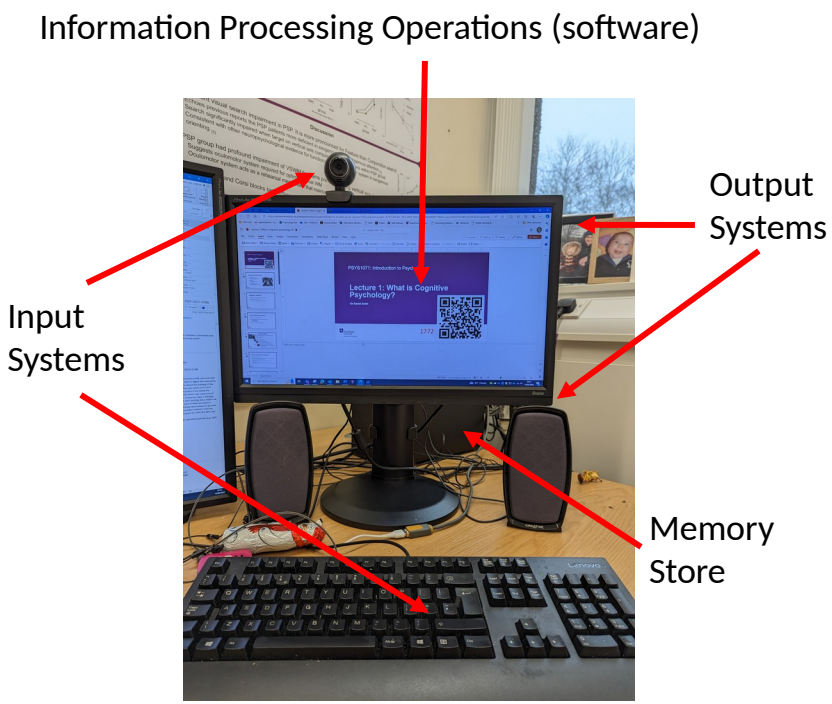
How **much** signal is needed to detect and identify a stimulus? (redundancy)

How **long** does it take to detect or identify a stimulus? (reaction time)

How **many** signals can be processed simultaneously? (capacity)

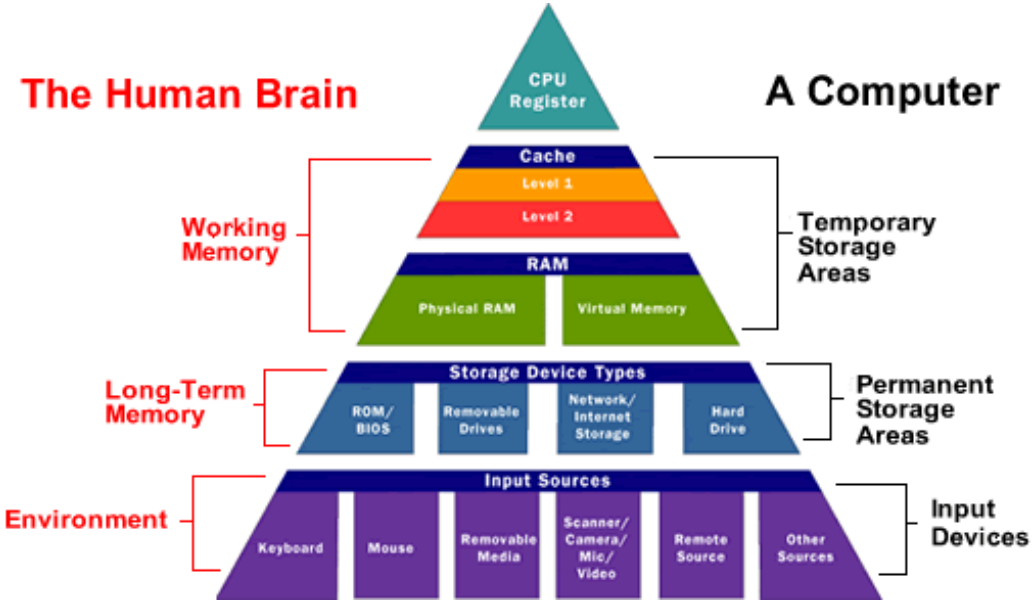
Allowed psychologists to operationalise and measure abstract concepts such as attention, memory, perception, planning, reasoning, motivation etc

The computational metaphor of the mind



- Brains are like computer hardware, cognition is software? (Searle 1994)
  - Sensory inputs transformed into an internal code ('internal representations')
  - Mental processes for processing these internal representations ('operations')
  - Limited capacity for running mental operations
  - Can store internal representations in a long-term memory and retrieve them for comparison with new inputs
  - Output via motor control system (speech, action)

The computational metaphor of the mind



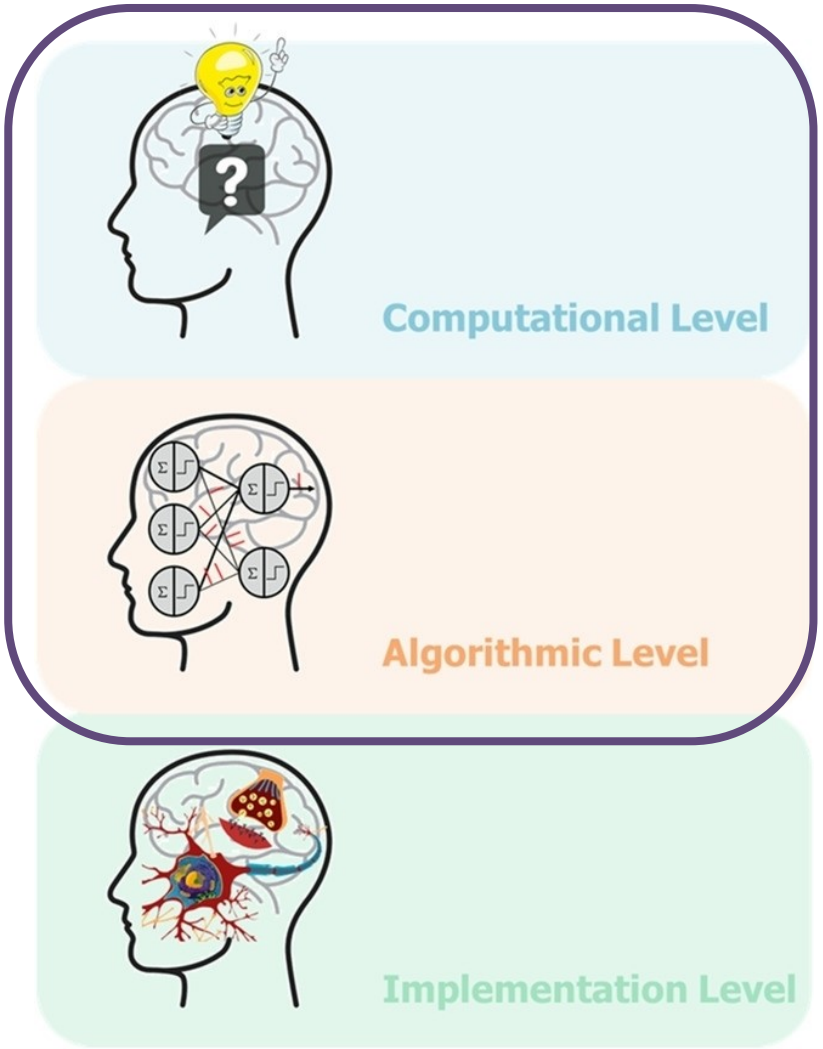
Assumptions

- The mind contains symbolic representations ('internal' or 'stimulus' representation)
  - Something that stands for, designates or represents something else
  - In computers, binary numbers symbolically represent specific pieces of information (e.g. a words, the colour of a pixel, a musical note).
  - There is a limited & well defined set of symbols
  - These symbolic representations are stored in memory
- Cognition is the product of 'operations'
  - Internal processes that act on symbolic representations
  - In computers, these are functions like copy, assign, insert, delete, save etc
  - In minds these operations might be 'recall', 'read', 'translate', 'store', etc etc
  - Operations are deployed according to rules that are also stored in memory
- Mental processes are like computer programmes that transform representations

3 Levels of Description (Marr 1982)

1. Computational Theory level
   - What is the function of cognition?
   - What is cognition for?
   - What cognitive functions are there?
2. Representation and Algorithm level
   - How does the cognition work?
   - How is information stored or “internally represented”
   - What operations or “algorithms” are used to manipulate internal representations?
3. Hardware level
   - How are the representations instantiated in the real world?

- Our primary interest as cognitive psychologists are levels 1 & 2



Marr D, 1982 *Vision. A Computational Investigation into the Human Representation and Processing of Visual Information* (New York: W H Freeman)

3 Levels of Description (Marr 1982)

1.<u>Computational Level (What and Why):</u>
The coffee maker's purpose is to produce a cup of coffee. It takes inputs like water, coffee grounds, and electricity, and outputs brewed coffee. The goal is to extract flavour compounds from coffee grounds using hot water.

<u>2. Algorithmic Level (How):</u>
The coffee maker follows a specific algorithm:

- Heat water to a specific temperature
- Pass hot water through coffee grounds
- Filter the resulting liquid
- Collect the brewed coffee in a pot or cup
  This level also includes representations like water level indicators and temperature controls.

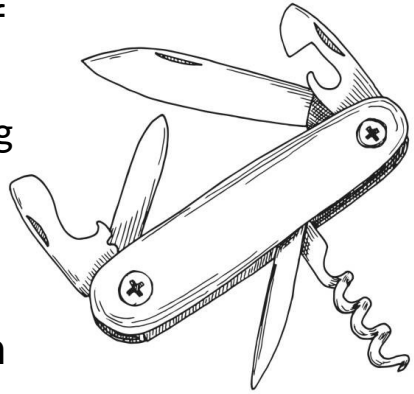
3.<u>Implementation Level (Physical Realization):</u>
This level describes the physical components that carry out the algorithm:

- Heating element to warm the water
- Pump to move water through the system
- Filter basket to hold coffee grounds
- Drip mechanism to control water flow
- Carafe to collect brewed coffee

Marr D, 1982 *Vision. A Computational Investigation into the Human Representation and Processing of Visual Information* (New York: W H Freeman)

Modularity I (Fodor, 1983)

- Human cognitions is organised into discrete mental modules, each of which fulfils a specific function:
  - Horizontal faculties: General competencies used across domains (e.g. long term memory)
  - Vertical faculties: Domain specific cognitive functions and processes (e.g. language production, object recognition)
- Modules don't need to interact with other psychological processes in order to operate (information encapsulation)
  - Disruption to one module does not necessarily affect processing in other modules
  - Damaging an object recognition module does not affect short-term memory
- Modules are similar across all humans
- Module processing is fast and obligatory (i.e. it happens without conscious thought and cannot be suppressed)
- Cognition involves the activation of horizontal and vertical faculties
  - E.g. naming a face might draw on a visual module, memory faculty to match visual representation to stored image and a linguistic module to produce the name



Fodor, Jerry A. (1983). The Modularity of Mind: An Essay on Faculty Psychology. Cambridge, MA: MIT Press.

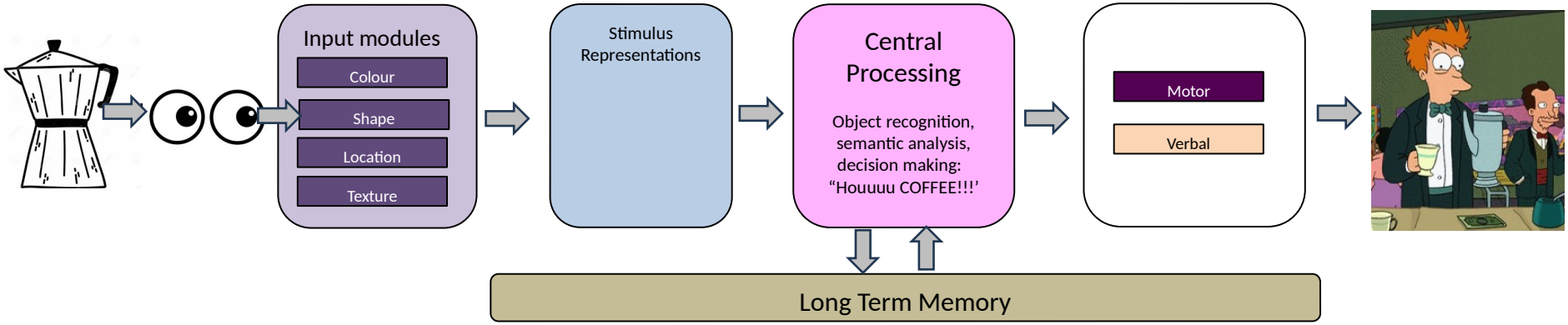
Modularity II

- Distinction between input systems *(typically these are vertical faculties)* and central processors

Input Systems
Process incoming sensory information
Transfer information to central processors
Domain specific (only process a particular class of information)

Central Processor
Makes decisions
Plans actions
Not modular

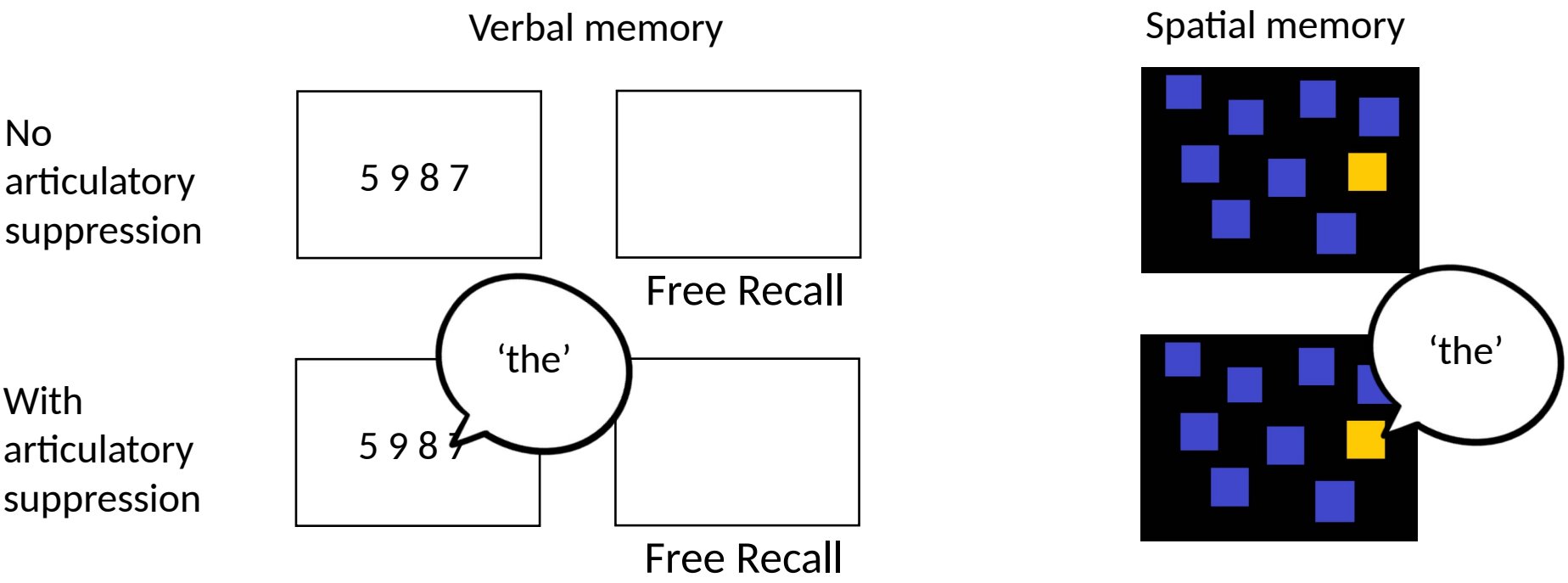
Effector Systems
Execute responses



Methods for identifying modules

How can we know how many modules there are?

- **<u>Dissociation:</u>** A manipulation that affects one cognitive task but not a different task



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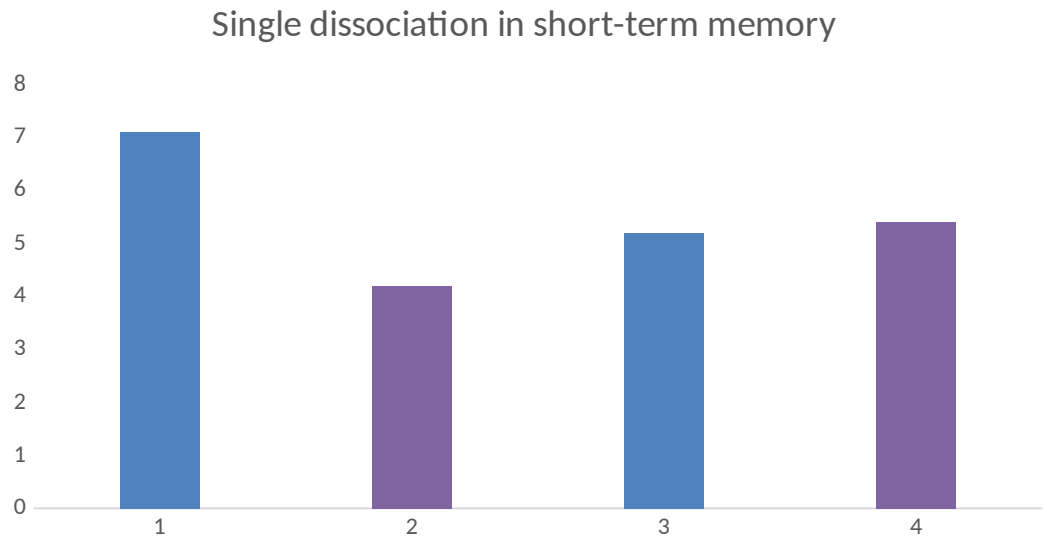

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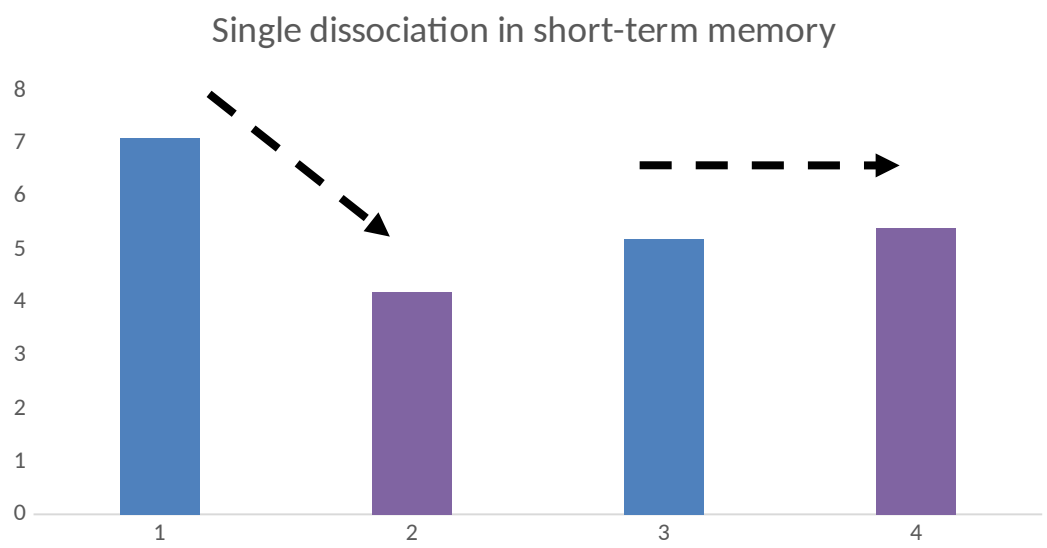

What do we see?

There is a decrease in performance for the phonological task when there is interference
However, the spatial task is not affected

What does it mean?

The interference is selectively impacting phonological (verbal) processing without affecting spatial processing
Phonological and spatial tasks likely rely on distinct cognitive subsystems within working memory

Methods for identifying modules

How can we know how many modules there are?

- **<u>Double Dissociation:</u>** two different cognitive functions are shown to be impaired independently of each other, demonstrating that they rely on separate brain processes or systems.

•Single dissociation: interference with module A impairs task X but not task Y.

Articulatory suppression (A) disrupts verbal (X) but not spatial memory (Y)

•Double dissociation: interference with A impairs task X but not task Y, while interference with B impairs task Y but not task X

Making saccades (eye-movements) (B) disrupts spatial(Y) but not verbal memory 9A)

Methods for identifying modules

How can we know how many modules there are?

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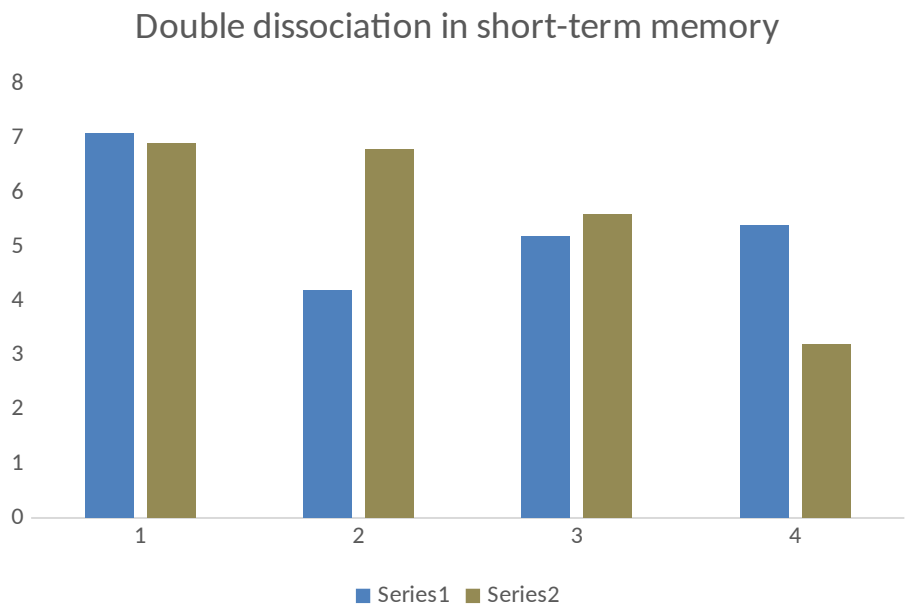

Methods for identifying modules

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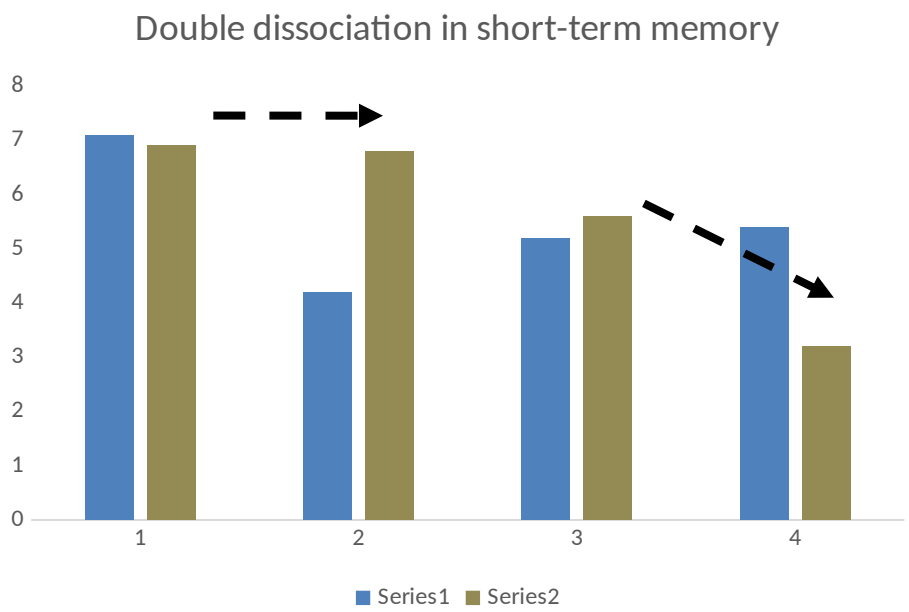


What do we see?

There is a decrease in performance for the spatial task when there is eye-movement interference but this time the phonological task is not affected

What does it mean?

The saccade task is specifically interfering with spatial processing while leaving phonological processing intact. Spatial and phonological tasks rely on separate cognitive systems

We finished the lecture on this… will catch up the 3 remaining slides on Friday 🏝️

Methods for identifying modules

Same applies for patient studies

	TASK	
	Digit Span	List Learning
LESION A	✓	X
LESION B	X	✓
CONTROL (no lesion)	✓	✓

Inference: there are processes/mechanisms involved in task X that are not involved in task Y and vice versa.

The two tasks are mediated by different functional mechanisms:

Task X controlled by area A

Task Y controlled by area B

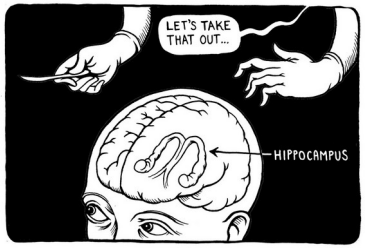
- Single dissociation allows researchers to infer that two functions are independent to some degree, but it doesn't rule out the possibility that they are partially related.

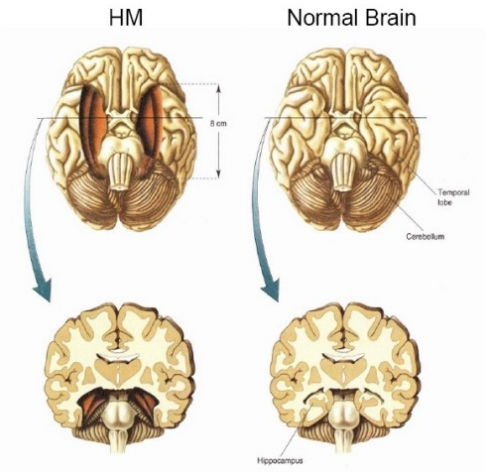
- Double dissociation enables more specific inferences about brain function and functional localization, conclusively demonstrating that two functions are separate

Cognitive Neuropsychology

- What can brain damage tell us about **normal** cognition?
  - Reverse engineering cognition
  - Localisation of functions less important
  - Typically investigate single cases

- Patient HM (Scoville & Milner 1957)
  - Neurosurgery to cure epilepsy
    - Severe anterograde amnesia: a couldn't form new memories
    - Short term memory OK, could learn new skills.
    - LTM, STM and Procedural memory **must** be different systems!
  - Revolutionised understanding of how memory functions





Scoville, W. B., & Milner, B. (1957). Loss of recent memory after bilateral hippocampal lesions. *Journal of neurology, neurosurgery, and psychiatry*, *20*(1), 11.

Limitations

- What is ‘normal’ performance for that patient?
  - Rarely know how good a person was at a task before injury

- **Plasticity & Functional reorganisation** of cognition, Neuroplasticity refers to the ability of the nervous system to adapt its structural organisation to new situations (e.g., after damage).
  - Perhaps patients adopt compensatory strategies

- Cannot say anything about time-course of information processing

- **Heterogeneity**: Damage is rarely focal, overlapping deficits/lesions, all patients are different!
  - E.g. Stroke affects lots of brain area and lots of cognitive systems

Summary

- Cognitive psychologists conceptualise thought as information processing
  - The goal is to understand these processes

- There are some key assumption about the nature of human cognition
  - Cognition is the consequence of information processing
  - Information processing comprises the storage and manipulation of internal representations according to well defined rules
  - Cognitive process are modular
  - Dissociations provide insight into the nature of these modules.

On Friday…

- The cognitive psychology of attention

After today's lecture you should be able to…

- Understand the aims of cognitive psychology

- Explain the assumptions and principles underpinning cognitive psychology
  - Information Processing analogy
  - The Computational metaphor
  - The principle of modular design

- Understand the idea of single and double dissociations

Test yourself!

- Why can we claim that spatial and phonological tasks rely on separate cognitive systems (or modules)?